

Initiating Search

November 5, 2025, 9:27 PM

• Search:

CAS Lexicon Search:

1,1'-Carbonyldiimidazole - Preferred Concept

Search Tasks

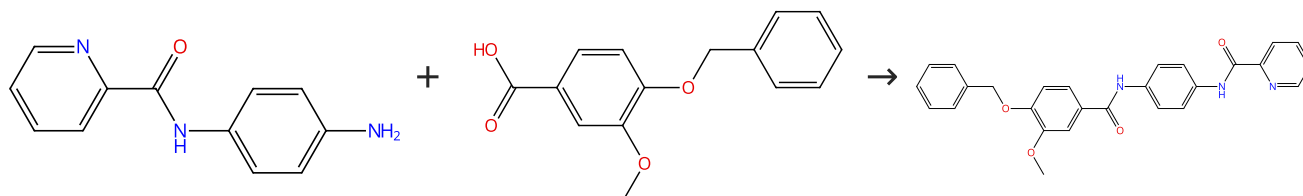
Task	Result Type	View
Returned Reference Results from CAS Lexicon (20,966)	 References	View Results
Exported: Retrieved Related Reaction Results + Filters (1,396)	 Reactions	View Results
Filtered By:		
Yield:	90-100%, 80-89%	
Document Type:	Journal	
Language:	English	
Publication Name:	Bioorganic & Medicinal Chemistry Letters, Bioorganic & Medicinal Chemistry, European Journal of Medicinal Chemistry, Journal of Medicinal Chemistry, Organic Process Research & Development	

Reactions (50)

[View in CAS SciFinder](#)

Scheme 1 (1 Reaction)

Steps: 1 Yield: 100%


 Suppliers (17)

 Suppliers (67)

31-367-CAS-12885868

Steps: 1 Yield: 100%

Rational design of inhibitors of the bacterial cell wall synthetic enzyme GlmU using virtual screening and lead-hopping

By: Doig, Peter; et al

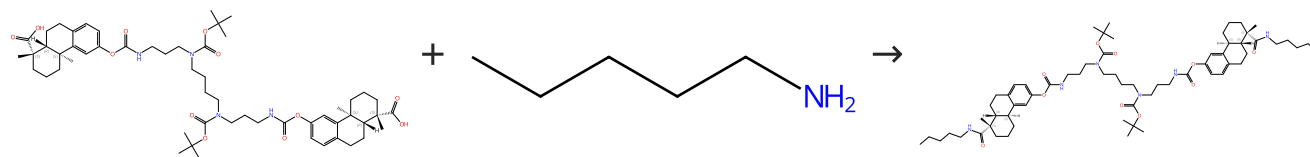
Bioorganic & Medicinal Chemistry (2014), 22(21), 6256-6269.

1.1 **Reagents:** 1-Hydroxybenzotriazole, Diisopropylethylamine, 1-[Bis(dimethylamino)methylene]-1*H*-benzotriazolium hexafluorophosphate(1-) 3-oxide
Solvents: Dimethylformamide; 1 d, rt

Experimental Protocols

Scheme 2 (1 Reaction)

Steps: 1 Yield: 100%

Absolute stereochemistry shown,
Rotation (+)
 Suppliers (55)
Absolute stereochemistry shown,
Rotation (+)

31-614-CAS-31966724

Steps: 1 Yield: 100%

Valorisation of the diterpene podocarpic acid - Antibiotic and antibiotic enhancing activities of polyamine conjugates

By: Li, Steven A.; et al

Bioorganic & Medicinal Chemistry (2022), 64, 116762.

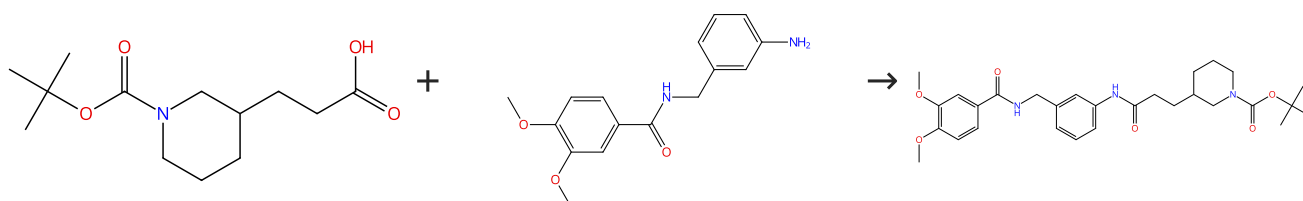
1.1 **Reagents:** Diisopropylethylamine, PyBOP
Solvents: Dimethylformamide; 5 h, rt

1.2 72 h, rt

Experimental Protocols

Scheme 3 (1 Reaction)

Steps: 1 Yield: 100%


 Suppliers (63)

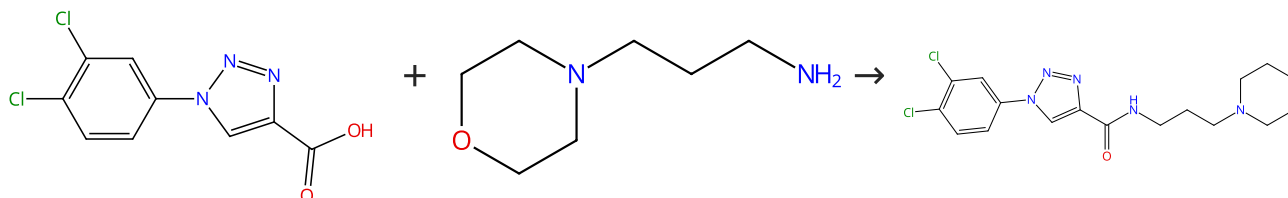
 Suppliers (3)

 Supplier (1)

31-367-CAS-6162229 Steps: 1 Yield: 100% 1.1 Reagents: 1<i>H</i>-Benzotriazole, 4-Methylmorpholine, 1-Ethyl-3-(3'-dimethylaminopropyl)carbodiimide hydrochloride Solvents: Dimethylformamide; 15 - 30 min, rt 1.2 16 - 70 h, rt Experimental Protocols	Hit to Lead Account of the Discovery of Bisbenzamide and Related Ureidobenzamide Inhibitors of Rho Kinase By: Morwick, Tina; et al Journal of Medicinal Chemistry (2010), 53(2), 759-777.
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Scheme 4 (1 Reaction)

Steps: 1 Yield: 100%



Suppliers (30)

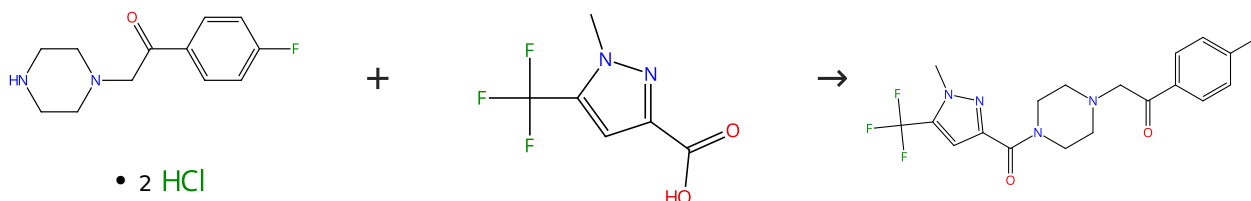
Suppliers (89)

Supplier (1)

31-367-CAS-23207899 Steps: 1 Yield: 100% 1.1 Reagents: 1,1'-Carbonyldiimidazole Solvents: Tetrahydrofuran; 100 min, reflux; reflux → rt 1.2 overnight, rt Experimental Protocols	Disrupting the Conserved Salt Bridge in the Trimerization of Influenza A Nucleoprotein By: Woodring, Jennifer L.; et al Journal of Medicinal Chemistry (2020), 63(1), 205-215.
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Scheme 5 (1 Reaction)

Steps: 1 Yield: 100%



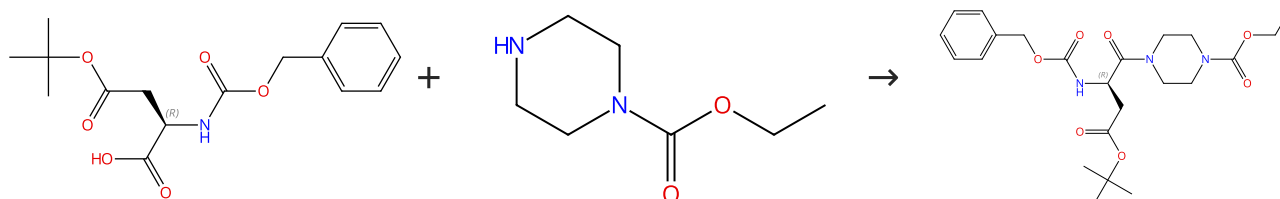
Suppliers (28)

Suppliers (63)

31-367-CAS-10608753 Steps: 1 Yield: 100% 1.1 Reagents: Triethylamine, <i>O</i>-(7-Azabenzotriazol-1-yl)-<i>N,N,N'</i>-tetramethyluronium hexafluorophosphate Solvents: Tetrahydrofuran; 10 min, 100 °C Experimental Protocols	Synthesis and in Vivo Evaluation of Phenethylpiperazine Amides: Selective 5-Hydroxytryptamine_{2A} Receptor Antagonists for the Treatment of Insomnia By: Xiong, Yifeng; et al Journal of Medicinal Chemistry (2010), 53(15), 5696-5706.
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Scheme 6 (1 Reaction)

Steps: 1 Yield: 100%



Absolute stereochemistry shown

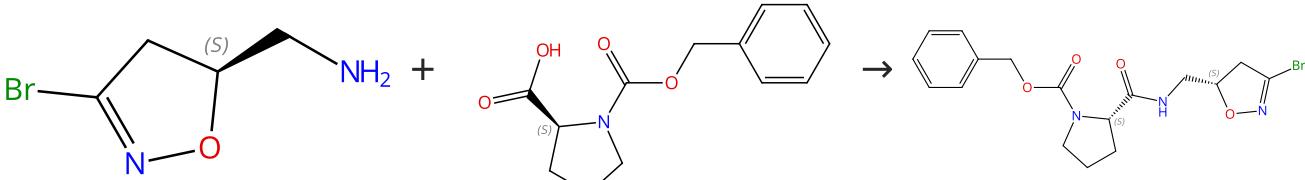
Suppliers (53)

Suppliers (78)

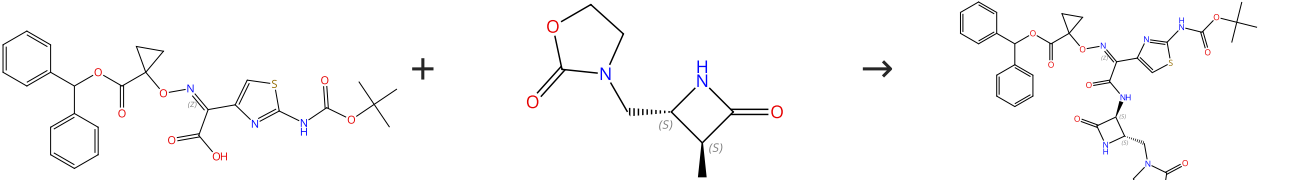
Absolute stereochemistry shown

Supplier (1)

31-367-CAS-13219160 Steps: 1 Yield: 100% 1.1 Reagents: 1-Hydroxybenzotriazole, Diisopropylethylamine, 1-Ethyl-3-(3'-dimethylaminopropyl)carbodiimide hydrochloride Solvents: Dichloromethane, Tetrahydrofuran; 15 min, rt 1.2 24 h, rt Experimental Protocols	4-((R)-2-[[6-((S)-3-Methoxypyrrolidin-1-yl)-2-phenylpyrimidine-4-carbonyl]amino]-3-phosphonopropionyl)piperazine-1-carboxylic Acid Butyl Ester (ACT-246475) and Its Prodrug (ACT-281959), a Novel P2Y₁₂ Receptor Antagonist with a Wider Therapeutic Window in the Rat Than Clopidogrel By: Caroff, Eva; et al Journal of Medicinal Chemistry (2015), 58(23), 9133-9153.
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Scheme 7 (1 Reaction)  Absolute stereochemistry shown Absolute stereochemistry shown, Rotation (-) Absolute stereochemistry shown Suppliers (2) Suppliers (98) Supplier (1)	Steps: 1 Yield: 100%
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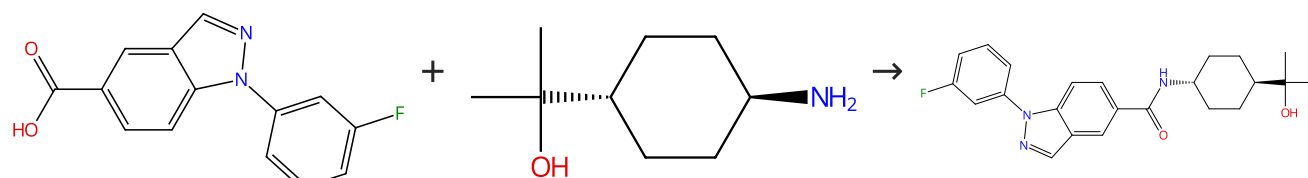
31-367-CAS-2558867 Steps: 1 Yield: 100% 1.1 Reagents: 4-Methylmorpholine, 1-Hydroxybenzotriazole, 1-Ethyl-3-(3'-dimethylaminopropyl)carbodiimide hydrochloride Solvents: Dimethylformamide; 30 min, rt Experimental Protocols	Discovery of Potent and Specific Dihydroisoxazole Inhibitors of Human Transglutaminase 2 By: Klock, Cornelius; et al Journal of Medicinal Chemistry (2014), 57(21), 9042-9064.
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Scheme 8 (1 Reaction)  Double bond geometry shown Absolute stereochemistry shown Relative stereochemistry shown Double bond geometry shown Suppliers (7) Supplier (1)	Steps: 1 Yield: 100%
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31-367-CAS-19020180 Steps: 1 Yield: 100% 1.1 Reagents: 4-Methylmorpholine, 2-Chloro-4,6-dimethoxy-1,3,5-triazine Solvents: Dimethylacetamide; 2 h, 0 - 5 °C 1.2 Solvents: Dimethylacetamide; 5 °C → rt; 20 h, rt 1.3 Solvents: Water; 30 min, rt Experimental Protocols	Optimization of novel monobactams with activity against carbapenem-resistant Enterobacteriaceae - Identification of L YS228 By: Reck, Folkert; et al Bioorganic & Medicinal Chemistry Letters (2018), 28(4), 748-755.
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Scheme 9 (1 Reaction)

Steps: 1 Yield: 100%



Suppliers (3)

Relative stereochemistry shown

Suppliers (60)

Relative stereochemistry shown

Supplier (1)

31-367-CAS-23964024

Steps: 1 Yield: 100%

1.1 **Reagents:** Diisopropylethylamine, *O*-(7-Azabenzotriazol-1-yl)-*N,N,N,N*-tetramethyluronium hexafluorophosphate
Solvents: Dimethylformamide

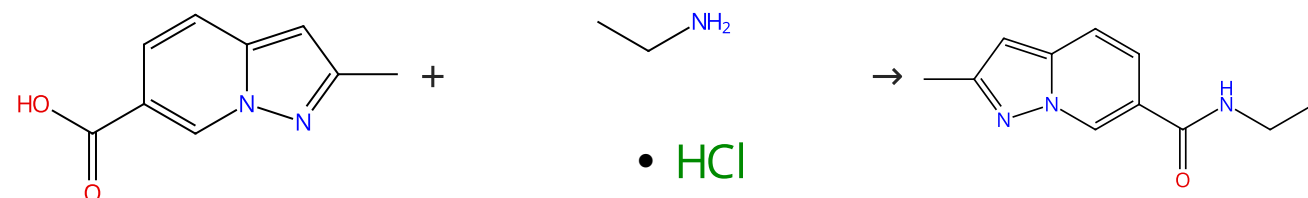
A knowledge-based, structural-aided discovery of a novel class of 2-phenylimidazo[1,2-*a*]pyridine-6-carboxamide H-PG DS inhibitors

By: Schulte, Christie A.; et al

Bioorganic & Medicinal Chemistry Letters (2021), 47, 128113.

Scheme 10 (1 Reaction)

Steps: 1 Yield: 100%



Suppliers (7)

Suppliers (47)

31-614-CAS-35314257

Steps: 1 Yield: 100%

1.1 **Reagents:** Diisopropylethylamine, *O*-(7-Azabenzotriazol-1-yl)-*N,N,N,N*-tetramethyluronium hexafluorophosphate
Solvents: Dimethylformamide; 16 h, rt

Discovery of High-Affinity Small-Molecule Binders of the Epigenetic Reader YEATS4

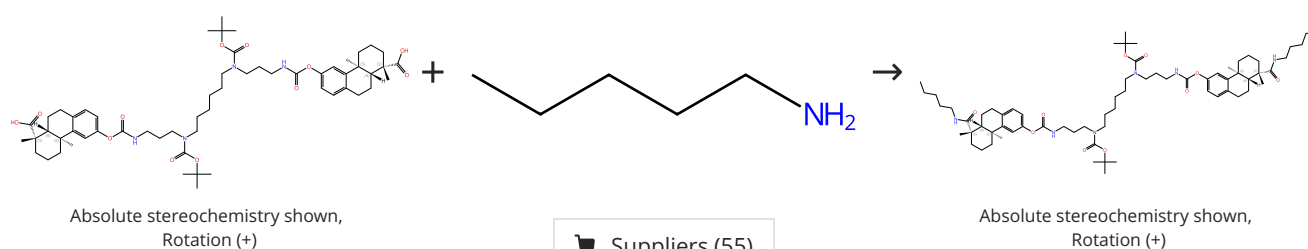
By: Londregan, Allyn T.; et al

Journal of Medicinal Chemistry (2023), 66(1), 460-472.

Experimental Protocols

Scheme 11 (1 Reaction)

Steps: 1 Yield: 100%

Absolute stereochemistry shown,
Rotation (+)

Suppliers (55)

Absolute stereochemistry shown,
Rotation (+)

31-614-CAS-31966733

Steps: 1 Yield: 100%

1.1 **Reagents:** Diisopropylethylamine, PyBOP
Solvents: Dimethylformamide; 5 h, rt

1.2 72 h, rt

Experimental Protocols

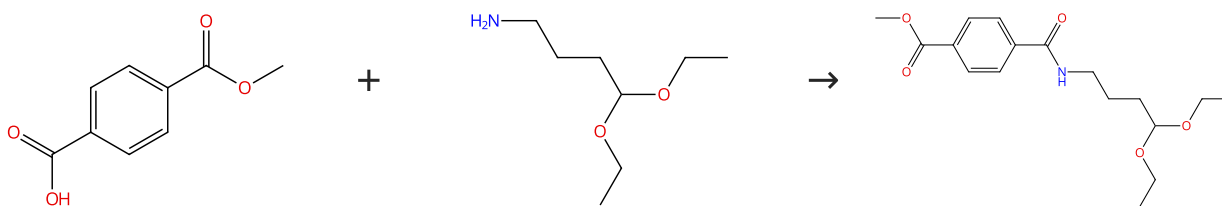
Valorisation of the diterpene podocarpic acid - Antibiotic and antibiotic enhancing activities of polyamine conjugates

By: Li, Steven A.; et al

Bioorganic & Medicinal Chemistry (2022), 64, 116762.

Scheme 12 (1 Reaction)

Steps: 1 Yield: 100%



Suppliers (89)

Suppliers (77)

31-614-CAS-41010363

Steps: 1 Yield: 100%

1.1 **Reagents:** Diisopropylethylamine, *O*-(7-Azabenzotriazol-1-yl)-*N,N,N,N*-tetramethyluronium hexafluorophosphate

Solvents: Dimethylformamide; 45 min, rt

1.2 18 h, rt

Experimental Protocols

Discovery, Multiparametric Optimization, and Solid-State Driven Identification of CHF-6550, a Novel Soft Dual Pharmacology Muscarinic Antagonist and Agonist (MABA) for the Inhaled Treatment of Respiratory Diseases

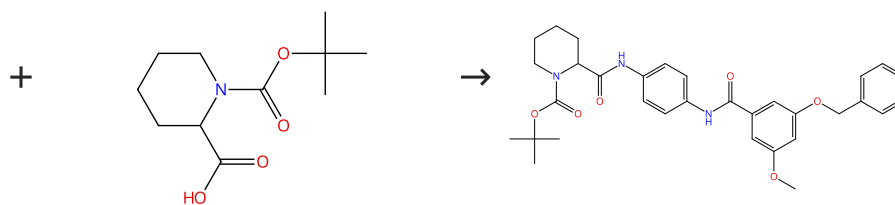
By: Carzaniga, Laura; et al

Journal of Medicinal Chemistry (2024), 67(12), 9816-9841.

Scheme 13 (1 Reaction)

Steps: 1 Yield: 100%

Multi-component
structure image
available in CAS
SciFinder



Suppliers (103)

31-367-CAS-10368334

Steps: 1 Yield: 100%

1.1 **Reagents:** 1-Hydroxybenzotriazole, Diisopropylethylamine, 1-[Bis(dimethylamino)methylene]-1*H*-benzotriazolium hexafluorophosphate(1-) 3-oxide

Solvents: Dimethylformamide; 3 d, rt

1.2 **Solvents:** Water; 30 min, rt

Experimental Protocols

Rational design of inhibitors of the bacterial cell wall synthetic enzyme GlmU using virtual screening and lead-hopping

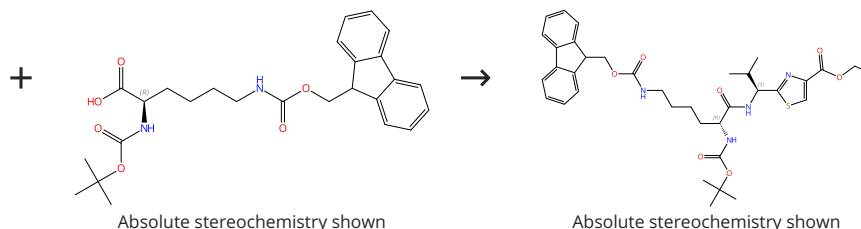
By: Doig, Peter; et al

Bioorganic & Medicinal Chemistry (2014), 22(21), 6256-6269.

Scheme 14 (1 Reaction)

Steps: 1 Yield: 100%

Multi-component
structure image
available in CAS
SciFinder



Suppliers (63)

Supplier (1)

Supplier (1)

31-367-CAS-23663323

Steps: 1 Yield: 100%

1.1 **Reagents:** Diisopropylethylamine, HOAt, *O*-(7-Azabenzotriazol-1-yl)-*N,N,N,N*-tetramethyluronium hexafluorophosphate

Solvents: Dimethylformamide; 3 h, rt

Experimental Protocols

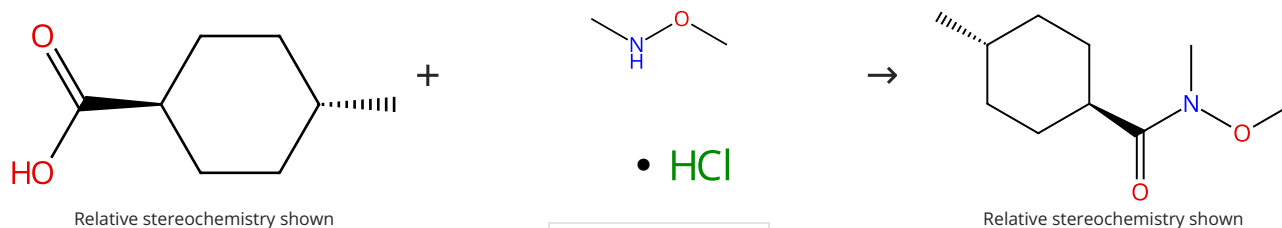
Automated design of macrocycles for therapeutic applications: From small molecules to peptides and proteins

By: Sindhikara, Dan; et al

Journal of Medicinal Chemistry (2020), 63(20), 12100-12115.

Scheme 15 (1 Reaction)

Steps: 1 Yield: 100%



Suppliers (70)

Suppliers (91)

Suppliers (8)

31-614-CAS-31541549

Steps: 1 Yield: 100%

Discovery of MK-4688: an Efficient Inhibitor of the HDM2-p53 Protein-Protein Interaction

By: Reutershan, Michael H.; et al

Journal of Medicinal Chemistry (2021), 64(21), 16213-16241.

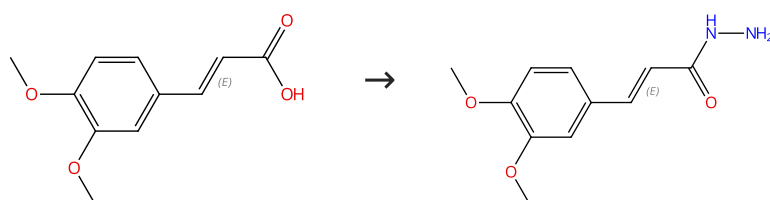
1.1 **Reagents:** Diisopropylethylamine, *O*-(7-Azabenzotriazol-1-yl)-*N,N,N,N*-tetramethyluronium hexafluorophosphate
Solvents: Dimethylformamide; 0.5 h, rt

1.2 rt; 3 h, rt

Experimental Protocols

Scheme 16 (1 Reaction)

Steps: 1 Yield: 100%



Suppliers (67)

Suppliers (3)

31-614-CAS-40592270

Steps: 1 Yield: 100%

Resveratrol-Based MTDLs to Stimulate Defensive and Regenerative Pathways and Block Early Events in Neurodegenerative Cascades

By: Herrera-Arozamena, Clara; et al

Journal of Medicinal Chemistry (2022), 65(6), 4727-4751.

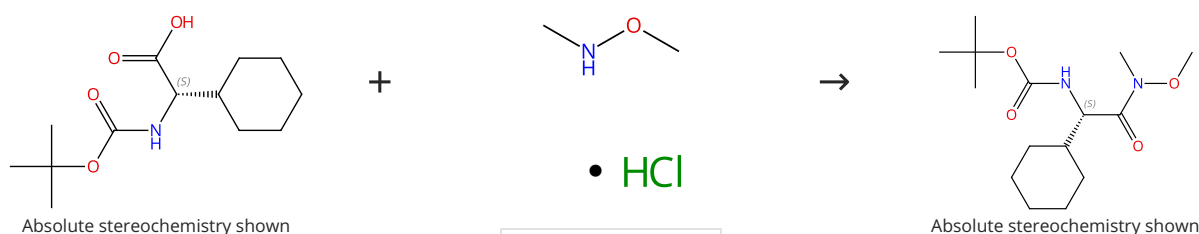
1.1 **Reagents:** 1-Hydroxybenzotriazole, 1-Ethyl-3-(3'-dimethylaminopropyl)carbodiimide hydrochloride
Catalysts: 4-(Dimethylamino)pyridine
Solvents: Acetonitrile; 30 - 180 min, rt

1.2 **Reagents:** Hydrazine hydrate (1:1); 1 - 15 min, rt

Experimental Protocols

Scheme 17 (1 Reaction)

Steps: 1 Yield: 100%



Suppliers (85)

Suppliers (91)

Suppliers (4)

31-367-CAS-5352417

Steps: 1 Yield: 100%

Depeptidization efforts on P3 - P2' α-ketoamide inhibitors of HCV NS3-4A serine protease: Effect on HCV replicon activity

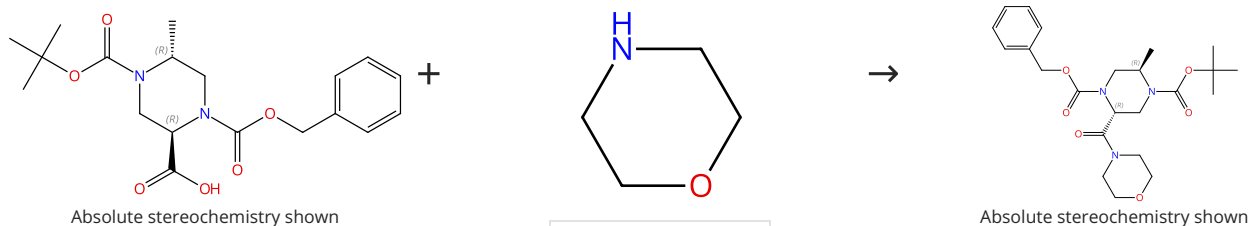
By: Bogen, Stephane L.; et al

Bioorganic & Medicinal Chemistry Letters (2006), 16(6), 1621-1627.

1.1 **Reagents:** 4-Methylmorpholine, 1-Ethyl-3-(3'-dimethylaminopropyl)carbodiimide hydrochloride
Solvents: Dichloromethane; -10 °C

Scheme 18 (1 Reaction)

Steps: 1 Yield: 100%



Absolute stereochemistry shown

Absolute stereochemistry shown

Suppliers (10)

Suppliers (88)

31-367-CAS-17327491

Steps: 1 Yield: 100%

Discovery of a Potent Nonpeptidomimetic, Small-Molecule Antagonist of Cellular Inhibitor of Apoptosis Protein 1 (cIAP1) and X-Linked Inhibitor of Apoptosis Protein (XIAP)

By: Tamanini, Emiliano; et al

Journal of Medicinal Chemistry (2017), 60(11), 4611-4625.

1.1 Reagents: 1,1'-Carbonyldiimidazole

Solvents: Dichloromethane; 3 h, rt

1.2 16 - 48 h, rt

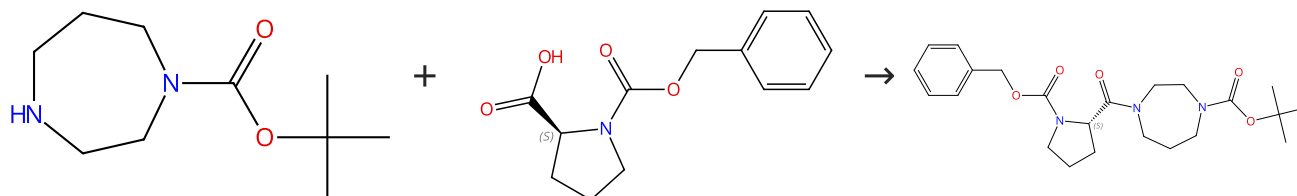
1.3 Reagents: Hydrochloric acid

Solvents: Water; 30 min, rt

Experimental Protocols

Scheme 19 (1 Reaction)

Steps: 1 Yield: 100%

Absolute stereochemistry shown,
Rotation (-)

Absolute stereochemistry shown

Suppliers (95)

Suppliers (98)

31-367-CAS-14572832

Steps: 1 Yield: 100%

Non-peptide oxytocin agonists

By: Pitt, Gary; et al

Bioorganic & Medicinal Chemistry Letters (2004), 14(17), 4585-4589.

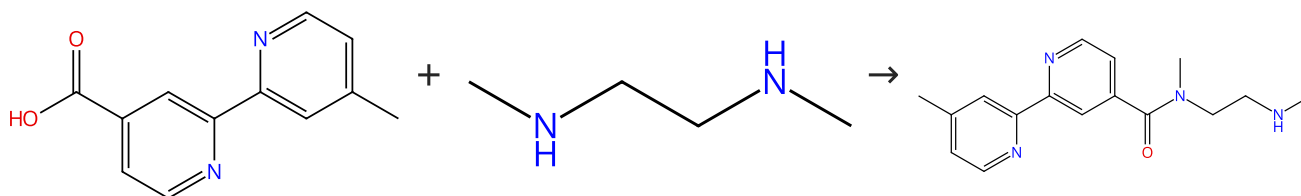
1.1 Reagents: Triethylamine, 1-Hydroxybenzotriazole, 1-Ethyl-3-

(3'-dimethylaminopropyl)carbodiimide hydrochloride

Solvents: Dichloromethane

Scheme 20 (1 Reaction)

Steps: 1 Yield: 100%



Suppliers (59)

Suppliers (83)

31-367-CAS-9908965

Steps: 1 Yield: 100%

DNA Interaction and photonic properties of DNA-Targeted acridine (2,2'-Bipyridine)Platinum (II) complexes

By: Gude, Lourdes; et al

Bioorganic & Medicinal Chemistry Letters (2002), 12(21), 3135-3139.

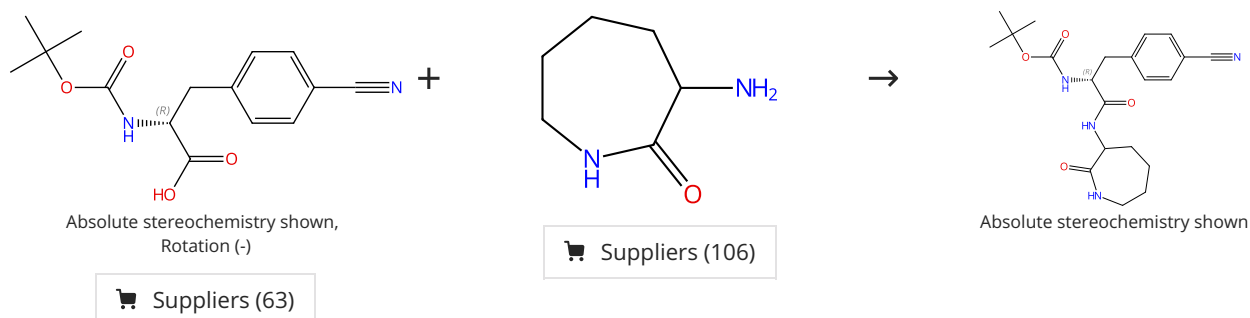
1.1 Reagents: 1,1'-Carbonyldiimidazole

Solvents: Tetrahydrofuran; 60 °C

1.2 Solvents: Tetrahydrofuran; rt

Scheme 21 (1 Reaction)

Steps: 1 Yield: 100%



31-614-CAS-40132282

Steps: 1 Yield: 100%

Design, Synthesis, Biological Evaluation, and Computational Studies of Novel Ureidopropanamides as Formyl Peptide Receptor 2 (FPR2) Agonists to Target the Resolution of Inflammation in Central Nervous System Disorders

- 1.1 Reagents: 1,1'-Carbonyldiimidazole
Solvents: Tetrahydrofuran; overnight, rt
- 1.2 24 h, rt

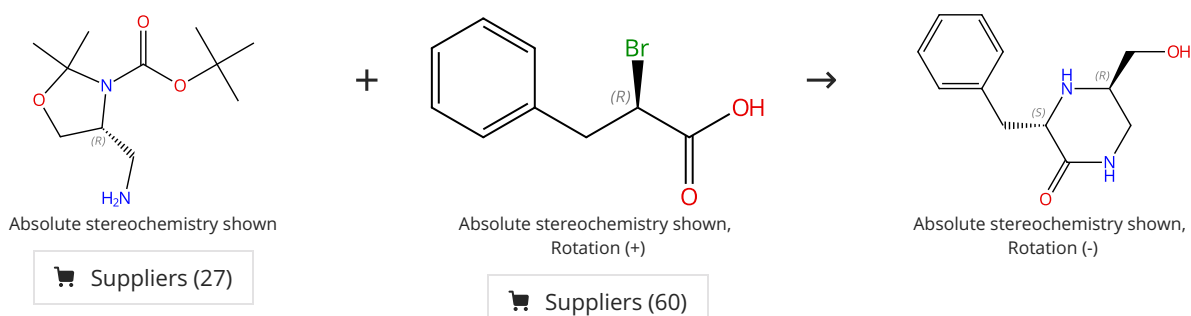
Experimental Protocols

By: Mastromarino, Margherita; et al

Journal of Medicinal Chemistry (2022), 65(6), 5004-5028.

Scheme 22 (1 Reaction)

Steps: 1 Yield: 100%



31-353-CAS-23853643

Steps: 1 Yield: 100%

Efficient synthesis of 5-(hydroxymethyl)piperazin-2-ones using automatically prepared chiral bromocarboxylic acid and Garner's aldehyde as versatile building blocks

- 1.1 Reagents: 1-Ethyl-3-(3'-dimethylaminopropyl)carbodiimide hydrochloride
Solvents: Dichloromethane; 10 h, rt
- 1.2 Reagents: Hydrochloric acid
Solvents: Water; rt
- 1.3 Reagents: Trifluoroacetic acid
Solvents: Dichloromethane; 1 h, rt
- 1.4 Reagents: Sodium carbonate
Solvents: Acetonitrile; 12 h, 50 °C

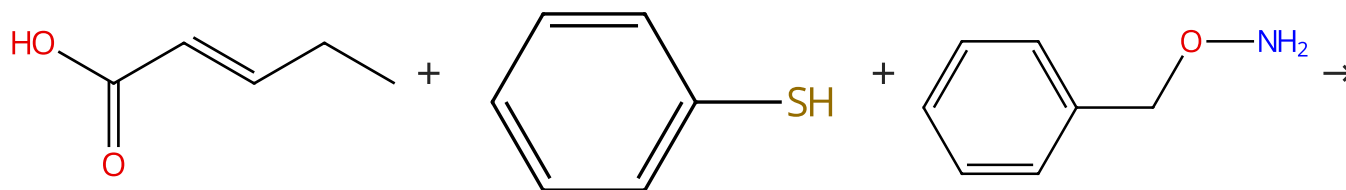
Experimental Protocols

By: Masui, Hisashi; et al

Bioorganic & Medicinal Chemistry Letters (2021), 40, 127961.

Scheme 23 (1 Reaction)

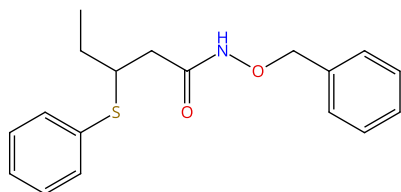
Steps: 1 Yield: 100%



Suppliers (12)

Suppliers (44)

Suppliers (68)



Supplier (1)

31-260-CAS-9425574

Steps: 1 Yield: 100%

Hydroxamic acid derivatives as potent peptide deformylase inhibitors and antibacterial agents

By: Apfel, Christian; et al

Journal of Medicinal Chemistry (2000), 43(12), 2324-2331.

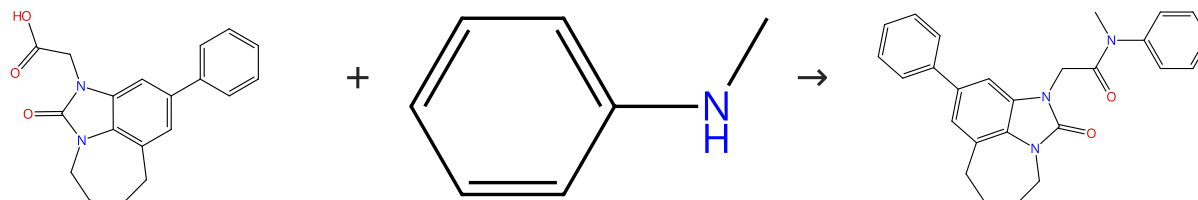
1.1 Reagents: Diisopropylethylamine
Solvents: Dimethylformamide; 48 h, 60 °C

1.2 Reagents: Dicyclohexylcarbodiimide, 1-Hydroxybenzotriazole
Solvents: Dichloromethane; overnight, rt

Experimental Protocols

Scheme 24 (1 Reaction)

Steps: 1 Yield: 100%



Suppliers (73)

Supplier (1)

31-367-CAS-2643456

Steps: 1 Yield: 100%

Design, synthesis and structure-activity relationship of novel tricyclic benzimidazolone derivatives as potent 18 kDa translocator protein (TSPO) ligands

By: Fukaya, Takayuki; et al

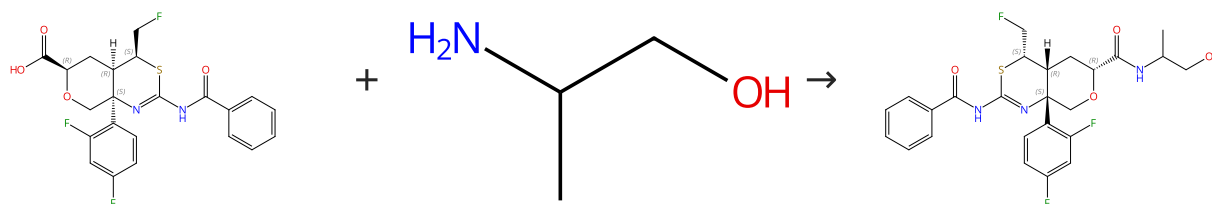
Bioorganic & Medicinal Chemistry (2013), 21(5), 1257-1267.

1.1 Reagents: 1-Hydroxybenzotriazole, 1-Ethyl-3-(3'-dimethylaminopropyl)carbodiimide hydrochloride
Solvents: Dimethylformamide; 7 h, rt

Experimental Protocols

Scheme 25 (1 Reaction)

Steps: 1 Yield: 100%



Absolute stereochemistry shown

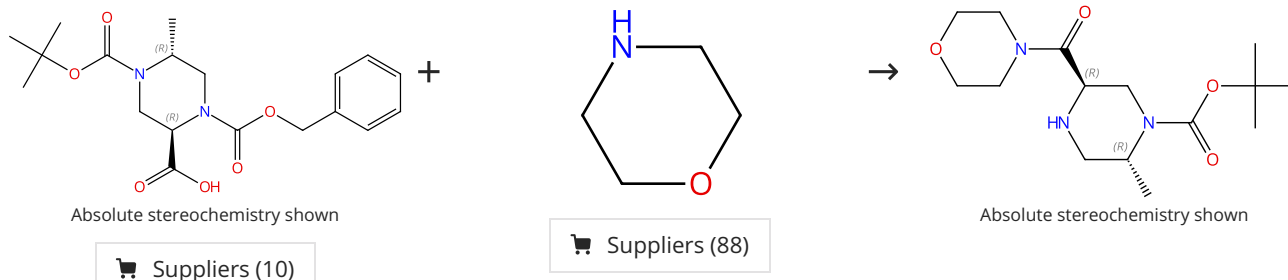
Absolute stereochemistry shown

Suppliers (78)

31-367-CAS-701541 Steps: 1 Yield: 100% 1.1 Reagents: Diisopropylethylamine, Methanaminium, <i>N</i>-[(dimethylamino)[(2-oxo-1(2<i>H</i>)-pyridinyl)oxy]methylene]-<i>N</i>-methyl-, tetrafluoroborate(1-) (1:1) Solvents: Dimethylformamide; 1 h, rt 1.2 3 h, rt Experimental Protocols	Utilizing Structures of CYP2D6 and BACE1 Complexes To Reduce Risk of Drug-Drug Interactions with a Novel Series of Centrally Efficacious BACE1 Inhibitors By: Brodney, Michael A.; et al Journal of Medicinal Chemistry (2015), 58(7), 3223-3252.
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Scheme 26 (1 Reaction)

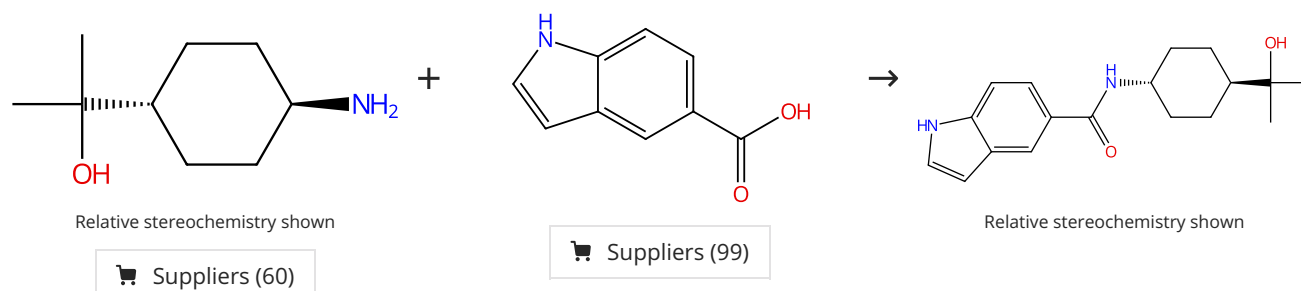
Steps: 1 Yield: 100%



31-353-CAS-17327495 Steps: 1 Yield: 100% 1.1 Reagents: 1,1'-Carbonyldiimidazole Solvents: Dichloromethane; 3 h, rt 1.2 16 - 48 h, rt 1.3 Reagents: Hydrochloric acid Solvents: Water; 30 min, rt 1.4 Reagents: Hydrogen Catalysts: Palladium Solvents: Methanol; 16 - 48 h, rt Experimental Protocols	Discovery of a Potent Nonpeptidomimetic, Small-Molecule Antagonist of Cellular Inhibitor of Apoptosis Protein 1 (cIAP1) and X-Linked Inhibitor of Apoptosis Protein (XIAP) By: Tamanini, Emiliano; et al Journal of Medicinal Chemistry (2017), 60(11), 4611-4625.
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Scheme 27 (1 Reaction)

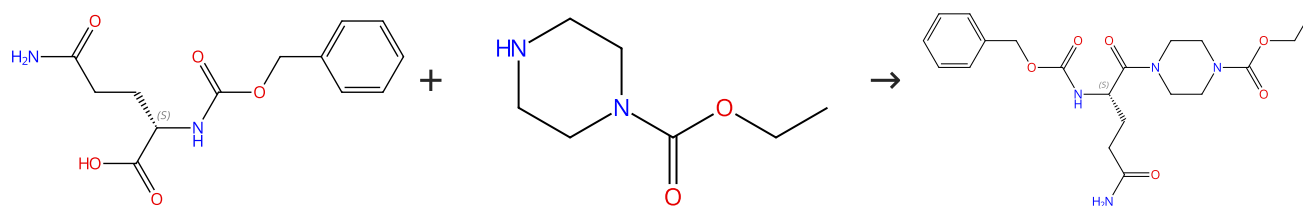
Steps: 1 Yield: 100%



31-614-CAS-23932493 Steps: 1 Yield: 100% 1.1 Reagents: Diisopropylethylamine, <i>O</i>-(7-Azabenzotriazol-1-yl)-<i>N,N,N,N</i>-tetramethyluronium hexafluorophosphate Solvents: Dimethylformamide	A knowledge-based, structural-aided discovery of a novel class of 2-phenylimidazo[1,2-<i>a</i>]pyridine-6-carboxamide H-PG DS inhibitors By: Schulte, Christie A.; et al Bioorganic & Medicinal Chemistry Letters (2021), 47, 128113.
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Scheme 28 (1 Reaction)

Steps: 1 Yield: 100%



Absolute stereochemistry shown

Suppliers (73)

Suppliers (78)

Absolute stereochemistry shown

Supplier (1)

31-367-CAS-10769811

Steps: 1 Yield: 100%

- 1.1 **Reagents:** 1-Hydroxybenzotriazole, 1-Ethyl-3-(3'-dimethylamino propyl)carbodiimide hydrochloride
Solvents: Dichloromethane; 5 min, rt
- 1.2 24 h, rt

Experimental Protocols

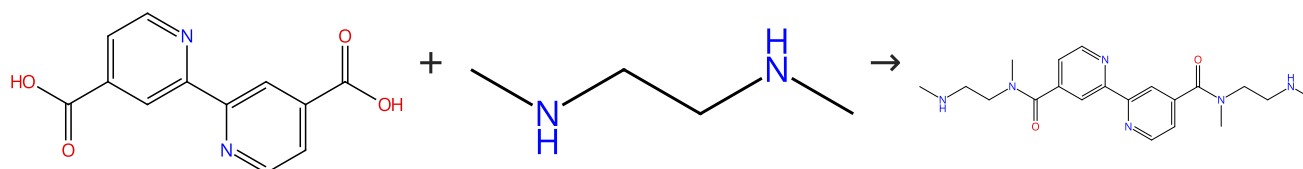
4-((R)-2-[[6-((S)-3-Methoxypyrrolidin-1-yl)-2-phenylpyrimidine-4-carbonyl]amino]-3-phosphonopropionyl)piperazine-1-carboxylic Acid Butyl Ester (ACT-246475) and Its Prodrug (ACT-281959), a Novel P2Y₁₂ Receptor Antagonist with a Wider Therapeutic Window in the Rat Than Clopidogrel

By: Caroff, Eva; et al

Journal of Medicinal Chemistry (2015), 58(23), 9133-9153.

Scheme 29 (1 Reaction)

Steps: 1 Yield: 100%



Suppliers (99)

Suppliers (83)

31-367-CAS-14175575

Steps: 1 Yield: 100%

- 1.1 **Reagents:** 1,1'-Carbonyldiimidazole
Solvents: Tetrahydrofuran; 60 °C
- 1.2 **Solvents:** Tetrahydrofuran; rt

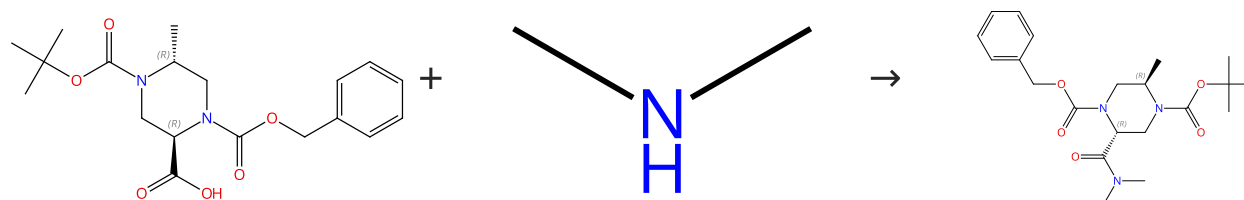
DNA Interaction and photonicking properties of DNA-Targeted acridine (2,2'-Bipyridine)Platinum (II) complexes

By: Gude, Lourdes; et al

Bioorganic & Medicinal Chemistry Letters (2002), 12(21), 3135-3139.

Scheme 30 (1 Reaction)

Steps: 1 Yield: 100%



Absolute stereochemistry shown

Suppliers (10)

Suppliers (118)

Absolute stereochemistry shown

31-367-CAS-17327490

Steps: 1 Yield: 100%

- 1.1 **Reagents:** 1,1'-Carbonyldiimidazole
Solvents: Dichloromethane; 3 h, rt
- 1.2 16 - 48 h, rt
- 1.3 **Reagents:** Hydrochloric acid
Solvents: Water; 30 min, rt

Experimental Protocols

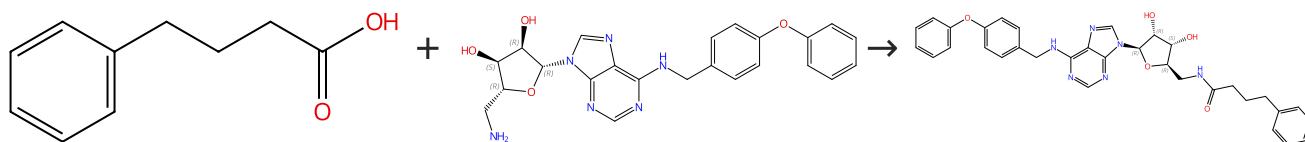
Discovery of a Potent Nonpeptidomimetic, Small-Molecule Antagonist of Cellular Inhibitor of Apoptosis Protein 1 (cIAP1) and X-Linked Inhibitor of Apoptosis Protein (XIAP)

By: Tamanini, Emiliano; et al

Journal of Medicinal Chemistry (2017), 60(11), 4611-4625.

Scheme 31 (1 Reaction)

Steps: 1 Yield: 100%



Absolute stereochemistry shown

Absolute stereochemistry shown

Suppliers (93)

Supplier (1)

31-367-CAS-10806846

Steps: 1 Yield: 100%

Inhibitors of adenosine consuming parasites through polymer-assisted solution phase synthesis of lipophilic 5'-amido-5'-deoxyadenosine derivatives

By: Heidler, Philipp; et al

Bioorganic & Medicinal Chemistry (2009), 17(4), 1428-1436.

1.1 **Reagents:** 2,3,5,6-Tetrafluoro-4-hydroxybenzamide (polymer-supported)

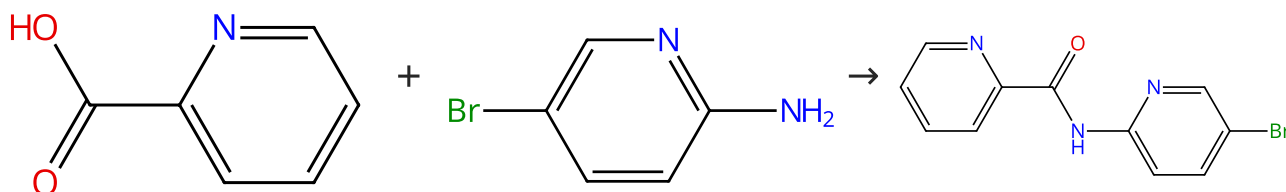
Solvents: Dichloromethane; rt

1.2 **Solvents:** Tetrahydrofuran; 12 h, rt

Experimental Protocols

Scheme 32 (1 Reaction)

Steps: 1 Yield: 100%



Suppliers (108)

Suppliers (100)

Suppliers (16)

31-367-CAS-20836489

Steps: 1 Yield: 100%

Synthesis and structure-bactericidal activity relationships of non-ketolides: 9-Oxime clarithromycin 11,12-cyclic carbonate featured with three-to eight-atom-length spacers at 3-OH

By: Li, Xue-Meng; et al

European Journal of Medicinal Chemistry (2019), 171, 235-254.

1.1 **Reagents:** Triethylamine, Pivaloyl chloride

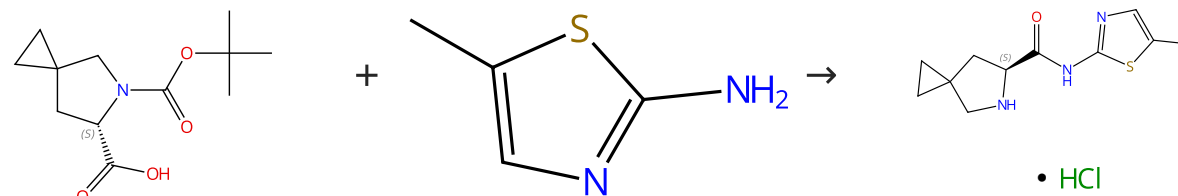
Solvents: Dichloromethane; 45 min, -15 °C

1.2 **Catalysts:** 4-(Dimethylamino)pyridine; 4.5 h, rt

Experimental Protocols

Scheme 33 (1 Reaction)

Steps: 1 Yield: 100%



Absolute stereochemistry shown,
Rotation (-)

Absolute stereochemistry shown

Suppliers (69)

Suppliers (99)

31-353-CAS-15730977

Steps: 1 Yield: 100%

New peptide deformylase inhibitors design, synthesis and pharmacokinetic assessment

By: Lv, Fengping; et al

Bioorganic & Medicinal Chemistry Letters (2016), 26(15), 3714-3718.

1.1 **Reagents:** Methanesulfonyl chloride, 1-Methylimidazole

Solvents: Dimethylformamide; rt → 0 °C; 0 °C; 15 min, 0 °C

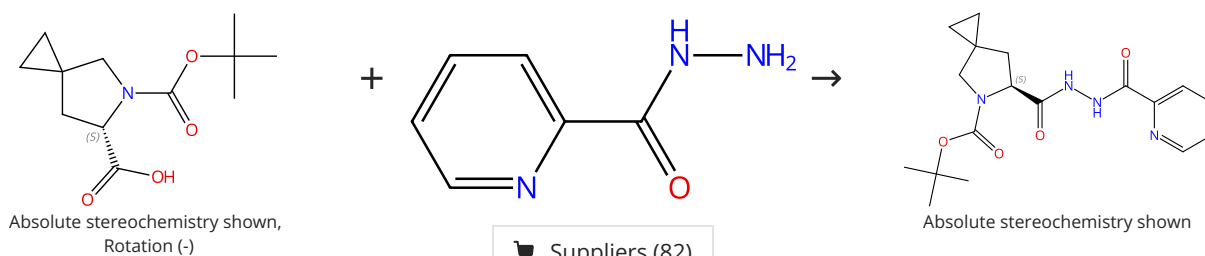
1.2 -

1.3 **Reagents:** Hydrochloric acid

Solvents: Diethyl ether, Methanol; overnight, rt

Scheme 34 (1 Reaction)

Steps: 1 Yield: 100%



Suppliers (69)

Suppliers (82)

31-367-CAS-15730996

Steps: 1 Yield: 100%

New peptide deformylase inhibitors design, synthesis and pharmacokinetic assessment

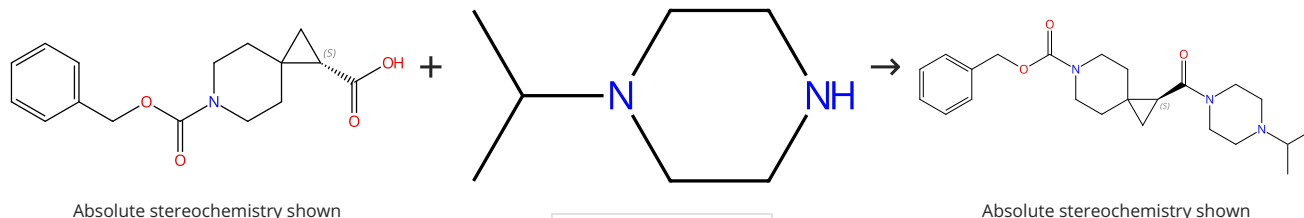
By: Lv, Fengping; et al

Bioorganic & Medicinal Chemistry Letters (2016), 26(15), 3714-3718.

- 1.1 **Reagents:** Methanesulfonyl chloride, 1-Methylimidazole
Solvents: Dimethylformamide; rt → 0 °C; 0 °C; 15 min, 0 °C
 1.2 6 h

Scheme 35 (1 Reaction)

Steps: 1 Yield: 100%



Suppliers (25)

Suppliers (86)

31-367-CAS-10558943

Steps: 1 Yield: 100%

Discovery of Spirofused Piperazine and Diazepane Amides as Selective Histamine-3 Antagonists with in Vivo Efficacy in a Mouse Model of Cognition

By: Brown, Dean G.; et al

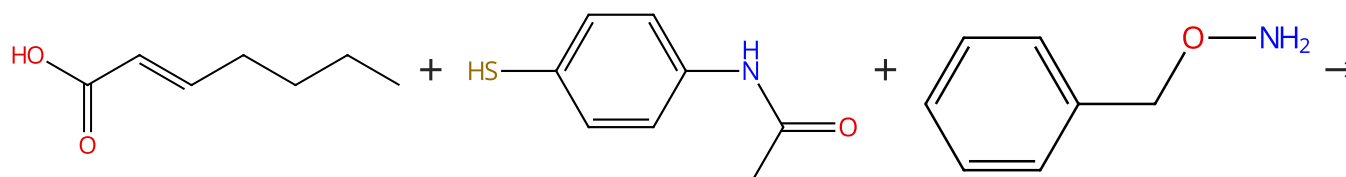
Journal of Medicinal Chemistry (2014), 57(3), 733-758.

- 1.1 **Reagents:** Diisopropylethylamine, 1-[Bis(dimethylamino)methylene]-1*H*-benzotriazolium hexafluorophosphate(1-) 3-oxide
Solvents: Dimethylformamide; 5 min, rt
 1.2 1 h, rt

Experimental Protocols

Scheme 36 (1 Reaction)

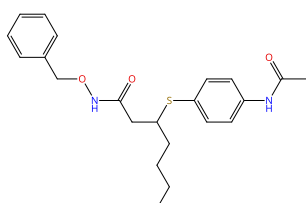
Steps: 1 Yield: 100%



Suppliers (62)

Suppliers (59)

Suppliers (68)

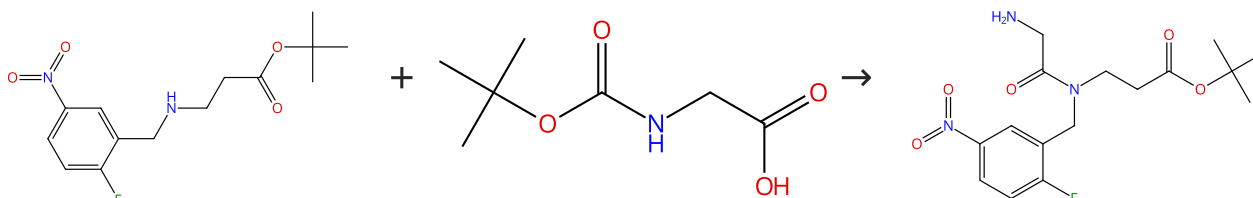


Supplier (1)

31-260-CAS-4890400 Steps: 1 Yield: 100% 1.1 Reagents: Diisopropylethylamine Solvents: Dimethylformamide; 48 h, 60 °C 1.2 Reagents: Dicyclohexylcarbodiimide, 1-Hydroxybenzotriazole Solvents: Dichloromethane; overnight, rt Experimental Protocols	Hydroxamic acid derivatives as potent peptide deformylase inhibitors and antibacterial agents By: Apfel, Christian; et al Journal of Medicinal Chemistry (2000), 43(12), 2324-2331.
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Scheme 37 (1 Reaction)

Steps: 1 Yield: 100%

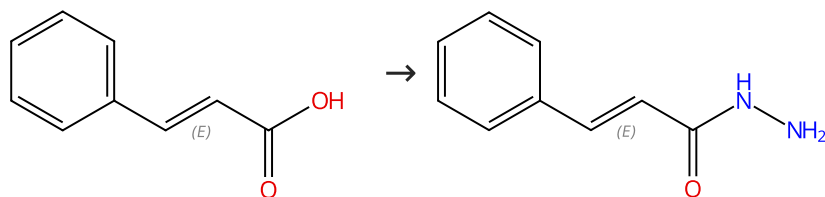


Suppliers (109)

31-353-CAS-8358937 Steps: 1 Yield: 100% 1.1 Reagents: Triethylamine, BOP reagent Solvents: Dichloromethane; overnight, rt 1.2 Reagents: Hydrochloric acid Solvents: Dichloromethane, 1,4-Dioxane; 3 h, 0 °C	Conformational Preferences in a Benzodiazepine Series of Potent Nonpeptide Fibrinogen Receptor Antagonists By: Keenan, Richard M.; et al Journal of Medicinal Chemistry (1999), 42(4), 545-559.
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Scheme 38 (1 Reaction)

Steps: 1 Yield: 100%



Double bond geometry shown

Double bond geometry shown

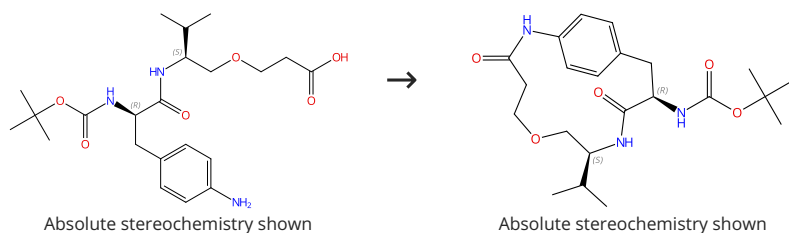
Suppliers (160)

Suppliers (12)

31-614-CAS-40592246 Steps: 1 Yield: 100% 1.1 Reagents: 1-Hydroxybenzotriazole, 1-Ethyl-3-(3'-dimethylaminopropyl)carbodiimide hydrochloride Catalysts: 4-(Dimethylamino)pyridine Solvents: Acetonitrile; 30 - 180 min, rt 1.2 Reagents: Hydrazine hydrate (1:1); 1 - 15 min, rt Experimental Protocols	Resveratrol-Based MTDLs to Stimulate Defensive and Regenerative Pathways and Block Early Events in Neurodegenerative Cascades By: Herrera-Arozamena, Clara; et al Journal of Medicinal Chemistry (2022), 65(6), 4727-4751.
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Scheme 39 (1 Reaction)

Steps: 1 Yield: 100%



Absolute stereochemistry shown

Absolute stereochemistry shown

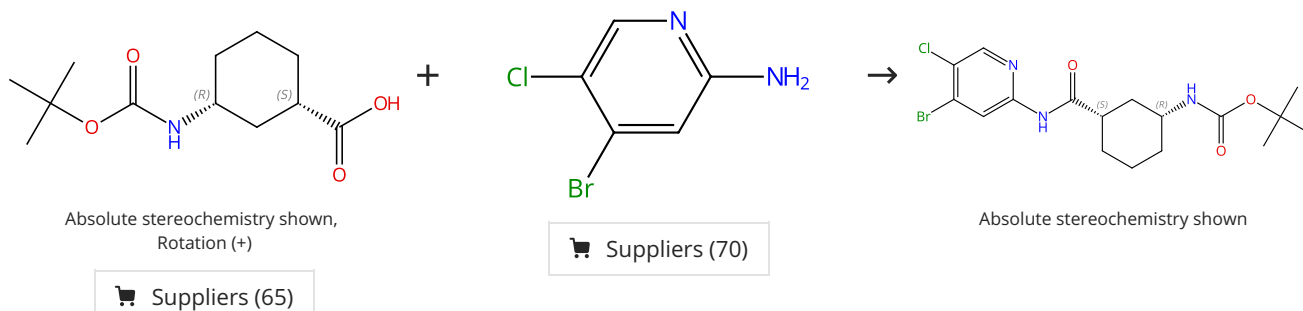
Supplier (1)

Supplier (1)

31-367-CAS-23659526 Steps: 1 Yield: 100% 1.1 Reagents: Diisopropylethylamine, HOAt, <i>O</i>-(7-Azabenzotriazol-1-yl)-<i>N,N,N,N</i>-tetramethyluronium hexafluoro phosphate Solvents: Dimethylformamide; 1 h, rt Experimental Protocols	Automated design of macrocycles for therapeutic applications: From small molecules to peptides and proteins By: Sindhikara, Dan; et al Journal of Medicinal Chemistry (2020), 63(20), 12100-12115.
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Scheme 40 (1 Reaction)

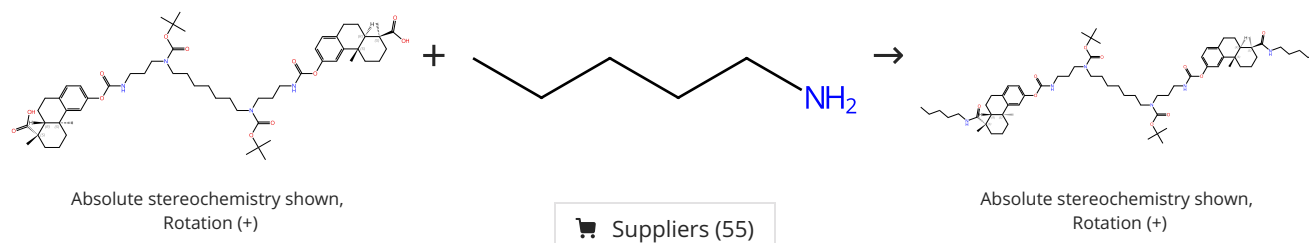
Steps: 1 Yield: 100%



31-614-CAS-31188286 Steps: 1 Yield: 100% 1.1 Reagents: 1-Chloro-<i>N,N</i>-2-trimethylpropenylamine Solvents: Dichloromethane; 0 °C; 1.5 h, rt 1.2 Reagents: Pyridine; 16 h, rt Experimental Protocols	Discovery of AZD4573, a Potent and Selective Inhibitor of C D K9 That Enables Short Duration of Target Engagement for the Treatment of Hematological Malignancies By: Barlaam, Bernard; et al Journal of Medicinal Chemistry (2020), 63(24), 15564-15590.
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Scheme 41 (1 Reaction)

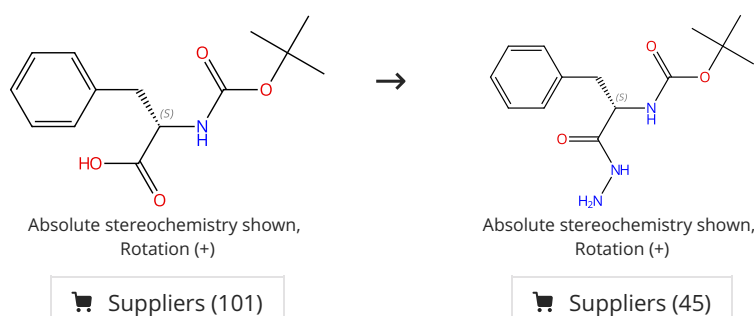
Steps: 1 Yield: 100%



31-614-CAS-31966731 Steps: 1 Yield: 100% 1.1 Reagents: Diisopropylethylamine, PyBOP Solvents: Dimethylformamide; 5 h, rt 1.2 72 h, rt Experimental Protocols	Valorisation of the diterpene podocarpic acid - Antibiotic and antibiotic enhancing activities of polyamine conjugates By: Li, Steven A.; et al Bioorganic & Medicinal Chemistry (2022), 64, 116762.
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Scheme 42 (1 Reaction)

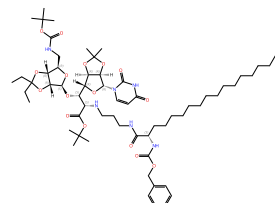
Steps: 1 Yield: 100%



31-367-CAS-9646621 Steps: 1 Yield: 100% 1.1 Reagents: Hydrazine, BOP reagent Solvents: Pyridine; 12 h, rt Experimental Protocols	Phenylimidazoles as Potent and Selective Inhibitors of Coagulation Factor XIa with in Vivo Antithrombotic Activity By: Hangeland, Jon J.; et al Journal of Medicinal Chemistry (2014), 57(23), 9915-9932.
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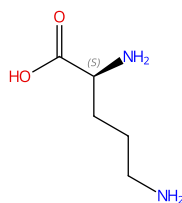
Scheme 43 (1 Reaction)

Steps: 1 Yield: 100%



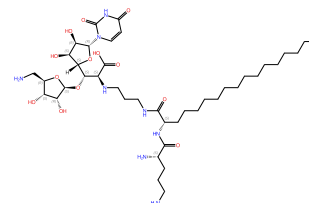
Absolute stereochemistry shown,
Rotation (+)

+



Absolute stereochemistry shown

→



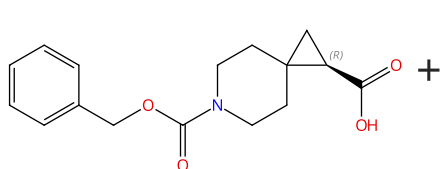
Absolute stereochemistry shown

Suppliers (57)

31-006-CAS-2739189 Steps: 1 Yield: 100% 1.1 Reagents: Hydrogen Catalysts: Palladium dihydroxide Solvents: Methanol; 5 h, rt 1.2 Reagents: <i>N</i> -Hydroxysuccinimide, 1-Ethyl-3-(3'-dimethylaminopropyl)carbodiimide hydrochloride Solvents: Dichloromethane; 1 h, 0 °C 1.3 Solvents: Dichloromethane; 0 °C; 8 h, rt 1.4 Reagents: Trifluoroacetic acid Solvents: Water; 8 h, rt Experimental Protocols	Mechanistic Analysis of Muraymycin Analogues: A Guide to the Design of MraY Inhibitors By: Tanino, Tetsuya; et al Journal of Medicinal Chemistry (2011), 54(24), 8421-8439.
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Scheme 44 (1 Reaction)

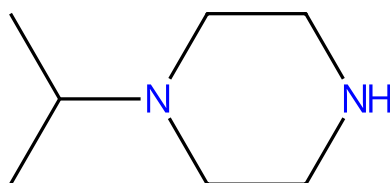
Steps: 1 Yield: 100%



Absolute stereochemistry shown

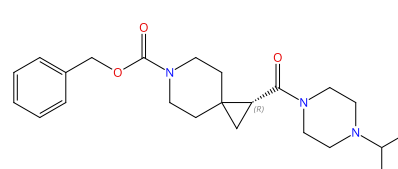
Suppliers (29)

+



Suppliers (86)

→



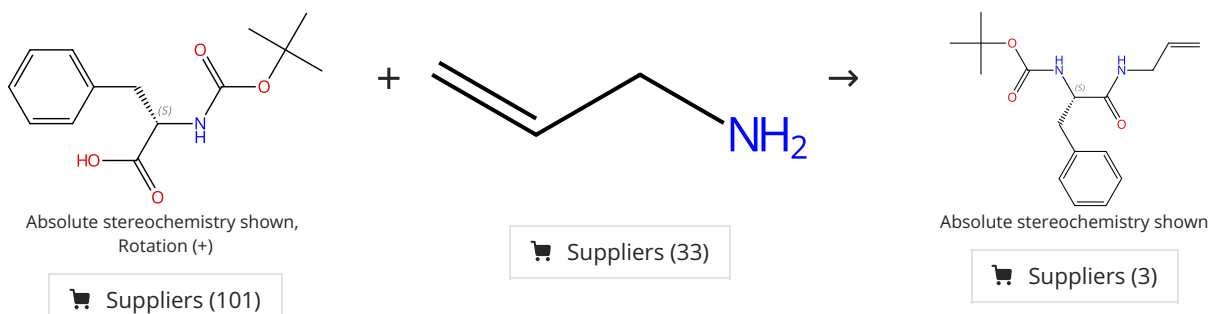
Absolute stereochemistry shown

Supplier (1)

31-367-CAS-583997 Steps: 1 Yield: 100% 1.1 Reagents: Diisopropylethylamine, 1-[Bis(dimethylamino)methylene]-1 <i>H</i> -benzotriazolium hexafluorophosphate(1-) 3-oxide Solvents: Dimethylformamide; 5 min, rt 1.2 1 h, rt Experimental Protocols	Discovery of Spirofused Piperazine and Diazepane Amides as Selective Histamine-3 Antagonists with in Vivo Efficacy in a Mouse Model of Cognition By: Brown, Dean G.; et al Journal of Medicinal Chemistry (2014), 57(3), 733-758.
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Scheme 45 (1 Reaction)

Steps: 1 Yield: 100%



31-367-CAS-2596466

Steps: 1 Yield: 100%

- 1.1 Reagents: 4-Methylmorpholine, Isobutyl chloroformate
Solvents: Tetrahydrofuran; rt → -25 °C; -25 °C → rt; 2 h, rt
- 1.2 Reagents: Sodium bisulfate
Solvents: Water; pH 1 - 2, rt

Experimental Protocols

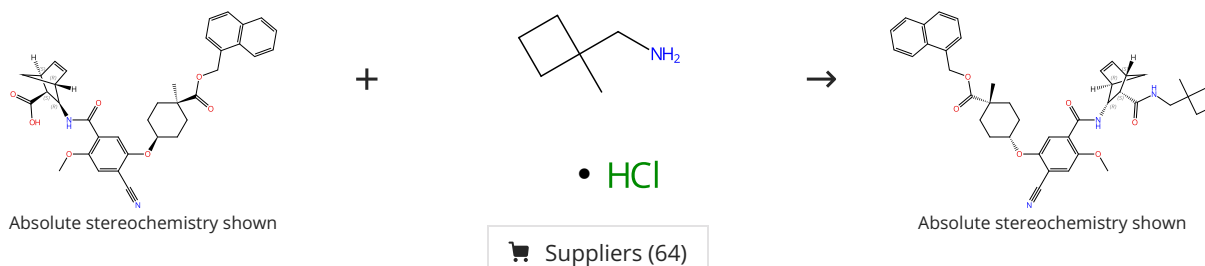
Interaction of Papain-like Cysteine Proteases with Dipeptide-Derived Nitriles

By: Loeser, Reik; et al

Journal of Medicinal Chemistry (2005), 48(24), 7688-7707.

Scheme 46 (1 Reaction)

Steps: 1 Yield: 100%



31-614-CAS-40332053

Steps: 1 Yield: 100%

- 1.1 Reagents: Diisopropylethylamine, *O*-(7-Azabenzotriazol-1-yl)-*N,N,N,N*-tetramethyluronium hexafluorophosphate
Solvents: Dimethylformamide; 30 min, rt

Experimental Protocols

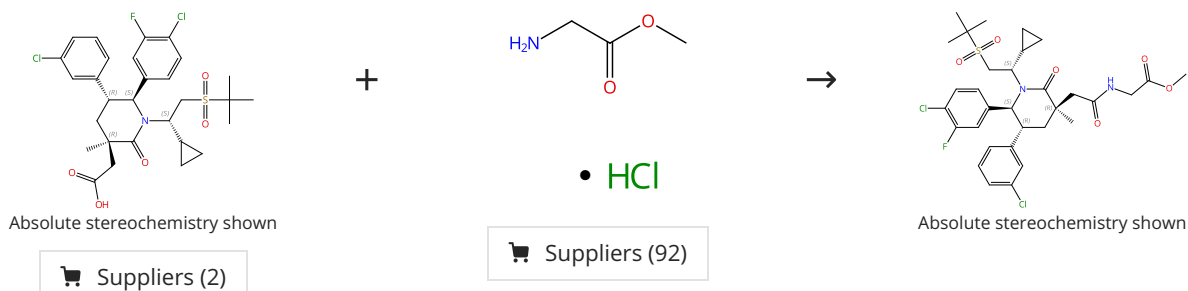
Discovery of Clinical Candidate AZD5462, a Selective Oral Allosteric RFXP1 Agonist for Treatment of Heart Failure

By: Granberg, Kenneth L.; et al

Journal of Medicinal Chemistry (2024), 67(6), 4419-4441.

Scheme 47 (1 Reaction)

Steps: 1 Yield: 100%



31-367-CAS-6605639

Steps: 1 Yield: 100%

- 1.1 Reagents: Sodium bicarbonate, 1-Ethyl-3-(3'-dimethylaminopropyl)carbodiimide, HOAt
Solvents: Dimethylformamide; rt; overnight, 40 °C

Experimental Protocols

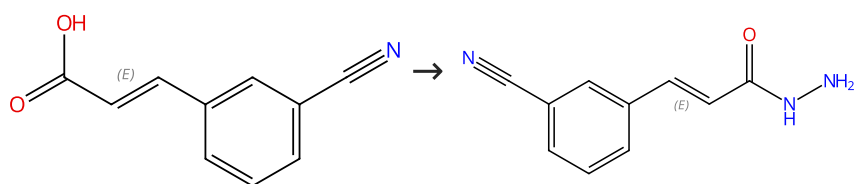
Novel Inhibitors of the MDM2-p53 Interaction Featuring Hydrogen Bond Acceptors as Carboxylic Acid Isosteres

By: Gonzalez, Ana Z.; et al

Journal of Medicinal Chemistry (2014), 57(7), 2963-2988.

Scheme 48 (1 Reaction)

Steps: 1 Yield: 100%



Double bond geometry shown

Double bond geometry shown

Suppliers (68)

Suppliers (2)

31-614-CAS-40592234

Steps: 1 Yield: 100%

- 1.1 **Reagents:** 1-Hydroxybenzotriazole, 1-Ethyl-3-(3'-dimethylaminopropyl)carbodiimide hydrochloride
Catalysts: 4-(Dimethylamino)pyridine
Solvents: Acetonitrile; 30 - 180 min, rt
- 1.2 **Reagents:** Hydrazine hydrate (1:1); 1 - 15 min, rt

Experimental Protocols

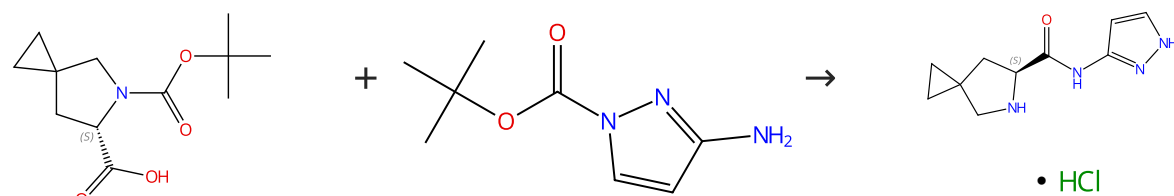
Resveratrol-Based MTDLs to Stimulate Defensive and Regenerative Pathways and Block Early Events in Neurodegenerative Cascades

By: Herrera-Arozamena, Clara; et al

Journal of Medicinal Chemistry (2022), 65(6), 4727-4751.

Scheme 49 (1 Reaction)

Steps: 1 Yield: 100%

Absolute stereochemistry shown,
Rotation (-)

Absolute stereochemistry shown

Suppliers (69)

Suppliers (59)

31-353-CAS-15730962

Steps: 1 Yield: 100%

- 1.1 **Reagents:** Methanesulfonyl chloride, 1-Methylimidazole
Solvents: Dimethylformamide; rt → 0 °C; 0 °C; 15 min, 0 °C
- 1.2 -
- 1.3 **Reagents:** Hydrochloric acid
Solvents: Diethyl ether, Methanol; overnight, rt

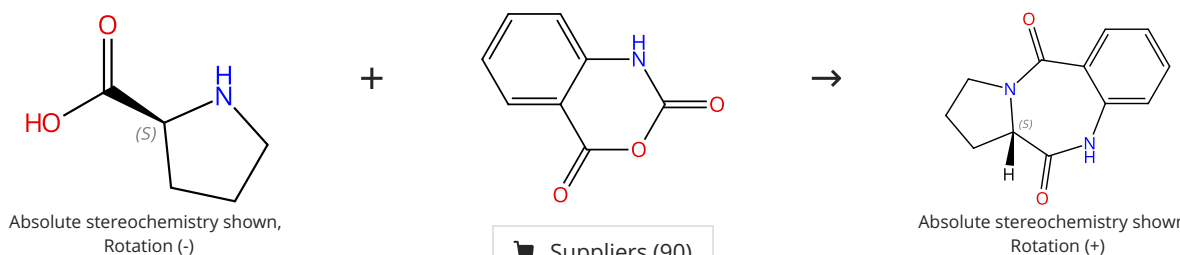
New peptide deformylase inhibitors design, synthesis and pharmacokinetic assessment

By: Lv, Fengping; et al

Bioorganic & Medicinal Chemistry Letters (2016), 26(15), 3714-3718.

Scheme 50 (1 Reaction)

Steps: 1 Yield: 100%

Absolute stereochemistry shown,
Rotation (-)Absolute stereochemistry shown,
Rotation (+)

Suppliers (168)

Suppliers (90)

Suppliers (35)

31-367-CAS-10659996

Steps: 1 Yield: 100%

- 1.1 **Solvents:** Dimethylformamide; 4 h, rt → 160 °C

Experimental Protocols

Synthesis and biological evaluation of oxadiazole derivatives as inhibitors of soluble guanylyl cyclase

By: von Wantoch Rekowski, Margarete; et al

Bioorganic & Medicinal Chemistry (2010), 18(3), 1288-1296.

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